

What is React Router → Q1

→ A react library that enables navigation b/w diff views/pages in a react App. without reloading the entire page.

Why we need it?

- React creates SPA's, mean only one HTML Page is loaded.
- For different page/views (Home, About, Contact)
- User can use browser back/forward buttons
- Each view should have its own URL (for bookmark)

Without React Router

```
const (page, setPage) = useState('home');
```

```
(page === 'home' && <Home />)
```

```
(page === 'about' && <About />)
```

With React Router

```
<Router>
```

```
<Route path="/" element={ <Home /> } />
```

```
<Route path="/about" element={ <About /> } />
```

```
</Routes>
```


BrowserRouter, Routes, Route Q3

■ BrowserRouter: wraps your entire app, tells React to use routing

■ Routes: A container for all your routes

■ Route: Define a path and which component to show

```
import { BrowserRouter, Routes, Route } from  
  "react-router-dom";
```

```
<BrowserRouter>
```

```
  <Routes>
```

```
    <Route path="/" element={ <Home /> } />
```

```
    <Route path="/about" element={ <About /> } />
```

```
  </Routes>
```

```
</BrowserRouter>
```

Q7

Link

NavLink

→ Basic navigation

→ No active styling

→ Use for buttons, cards

→ Used for navigation like <a> but without refreshing

→ Navigation with active state

→ Automatic active class/style

→ Use for navigation menus

→ Same as link but adds active class/style when link matches current URL

useNavigate Hook (Programmatic Navigation)

Q4

- redirect using JavaScript (like history.push)
- Redirect after form submission
- Navigate after authentication
- Go back / forward programmatically
- Navigate with state data

```
import { useNavigate } from "react-router-dom";  
const navigate = useNavigate();  
<button onClick = { () => navigate("/about") }  
Go to About </button>
```

Nested Routes

Q5

- Route inside another route like in Dashboard (Profile, Setting)

```
<Route path = "/dashboard" element = { <Dashboard  
  Route path = "profile" element = { <Profile /> } />  
  Route path = "settings" element = { <Setting /> } />  
</Route>
```

Route Parameter (useParams)

Q6

- Dynamic Segment in URLs
- Capture values from URLs


```
import { useParams } from 'react-router-dom';
```

// Route definition

```
<Route path="/user/:id" element={ <Profile /> } />
```

```
function Profile() {
```

```
  const { id } = useParams();
```

```
  return <h1>User ID: { id } </h1>
}
```

// Visiting /user/123 → displays "User ID: 123"

Query Parameters (useSearchParams)

→ Value after ? in URLs (e.g. ?category=fruit) are URL parameters

```
let [searchParams] = useSearchParams();
```

```
const category = searchParams.get("category")
```

useLocation

→ get current URL info (pathname, search, hash)

```
let location = useLocation();
```

```
console.log(location.pathname);
```


Protected Route Q8

→ A route that only show if user is logged in.

→ Restricted access to certain routes

```
function ProtectedRoute({ children }) {
```

```
  const isAuthenticated = false;
```

```
  return isAuthenticated ? children :
```

```
}
```

```
  <Navigate to="/Login"/>
```